LANDING TECHNIQUES

Landing was a tricky maneuver and much harder to master than taking off. A smooth touchdown required the pilot to adopt a reasonable angle of descent and switch off the engine at the right moment to reach the ground at a suitable landing speed. (More experienced pilots would switch a rotary engine on and off several times in their descent as a way of reducing speed.) Old hands often enjoyed watching novices come down, either landing too steep and too fast and bouncing alarmingly across the field, or losing speed too soon and "pancaking," often with spectacularly destructive effect on the undercarriage.

PITCHING FORKS

Curved skids on the front of this 1909 Deperdussin monoplane helped to stop the airplane pitching forward when landing on soft ground – a common hazard in the early days of flying.

learn. Although some aircraft were built with dual controls, most flight training took place with the trainee in an aircraft on his or her own. The lessons began with sessions taxiing on the ground. The trainee would trundle backward and forward across the airfield, turned around at each end by a mechanic. This got them used to the "noisy, jarring vibration of the engine" and the castor oil spraying back from the rotary engine into the pilot's face. This was followed by flying in short hops a few feet off the ground before the great moment of taking to the air. Fortunately the main aircraft of the time were quite stable and fairly easy to fly in good weather as long as they did not attempt any maneuvers. Grahame-White, writing in 1911, described the ease of maintaining level flight on a calm day: "One's feet move just a little to and fro upon the rudder bar. This little 'joggling' of the rudder is sufficient to keep the machine on a straight course. As regards the elevator, one is moving the rod in one's hand a matter of an inch or so…"

> "The danger? But danger is one of the attractions of flight."
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> **Jean Conneau**
> French aviator, 1911

Carelessness, foolhardiness, and inadequate landing grounds were all common causes of crashes. Early airplanes were fragile machines. Engine failure was common but did not necessarily lead to a crash. Airplanes could glide well, and an experienced pilot would expect to be able to nurse a powerless machine to a forced landing in a flat field. Structural failure, however, was a serious matter. If the wings or control surfaces collapsed under the pressure of sudden maneuvers or through the cumulative strain of use, a pilot was doomed. English car manufacturer Charles Rolls, for example, was killed in July 1910 when a rear elevator on his Wright biplane cracked as he came in to land, sending the machine diving into the ground.

Of course, much of the attraction of the early air shows was a ghoulish expectation of witnessing violent death. A journalist described the end of popular pilot Arch Hoxey, who lost control during an exhibition flight at the Los Angeles air meet in 1910: "The cracking of the spars and ripping of the cloth could be heard as the machine… came hurtling to the ground in a series of somersaults. When the attendants rushed to the tangled mass of wreckage they found the body crushed out of all semblance to a human being. The crowd waited until the announcer megaphoned the fatal news and then turned homeward." It had presumably been worth the price of the ticket.